Learned Target Assignment for Synthetic Tabular Data

Relabeling synthetic data to outperform real training data.

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Why generate *X* and *y* separately?

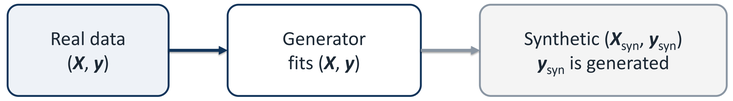
A supervised ML problem has input features ***X*** and a target ***y***. Synthetic data is only useful if it keeps the relationship between ***X*** and ***y***.

CTGAN, **TVAE** and **TabDDPM** model (***X***, ***y***) jointly, unaware of the target: ***y*** is one more feature column, and nothing ties a generated ***y*** to its ***X***.

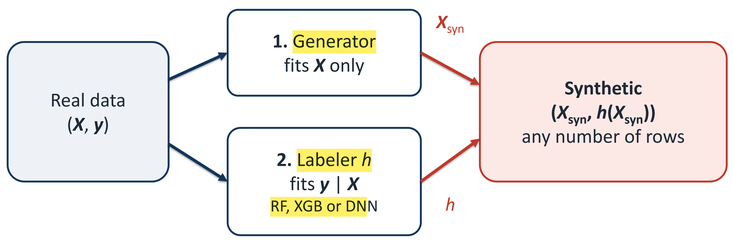
LTA is target-aware: it generates ***X***, then labels each row with a model fitted to the real data.

Method

Standard: joint synthesis



LTA: learned target assignment



- The generator samples synthetic features $\boldsymbol{X}_{syn}$ (100,000 rows here).
- The labeler h learns ***y*** from ***X*** on the real training rows.
- Ablation: keep a joint generator's $\boldsymbol{X}_{syn}$ and replace its $\boldsymbol{y}_{syn}$ with $h(\boldsymbol{X}_{syn})$.

Evaluation

- **Classification:** Adult, Census KDD, Covertype, MNIST12, MNIST28 (macro F1).
- **Regression:** California Housing, News (R^2).
- A downstream neural network trains on **synthetic only** or **real + synthetic**, and is tested on held-out real rows.
- Generators, labelers and preprocessing see training rows only; checkpoints are chosen on a real development split. 2 seeds; 588 downstream runs.

Takeaways

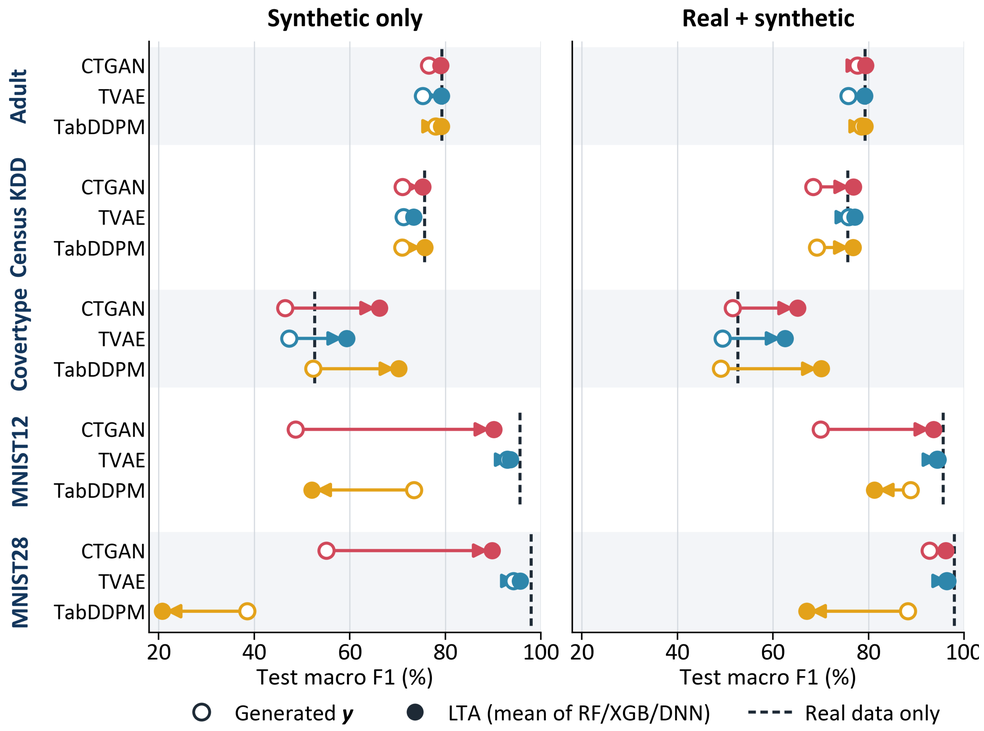
- Labeling synthetic features with a model trained on real data is a simple add-on to any generator.
- Exception: TabDDPM on MNIST. Its digits drift from real ones, so a real-data labeler mislabels them.
- **Caveats:** the wins over real data likely distill a strong labeler into many rows, and the real-only network is weak on Housing. With 5 tasks, the primary test (+7.4 points, p = 0.047) is not significant after Holm correction (p = 0.19).

Result: synthetic data that rivals real data

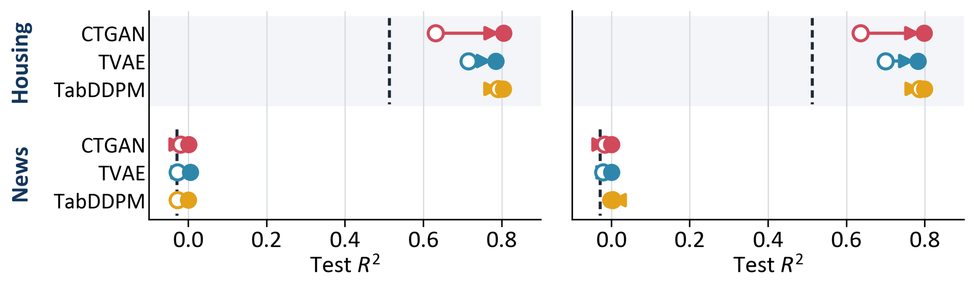
Synthetic-only training, classification: **LTA adds +7.2 macro-F1 points over generated targets**

95% CI 0.3 to 14.0; better in 13 of 15 task–generator pairs; CTGAN and TVAE: +12.3 points (18% relative).

With TVAE features, LTA comes within **2.3 points** of real-data training on every classification task, and **beats it** on Covertype (+6.7) and Housing (R^2 0.78 vs. 0.51).

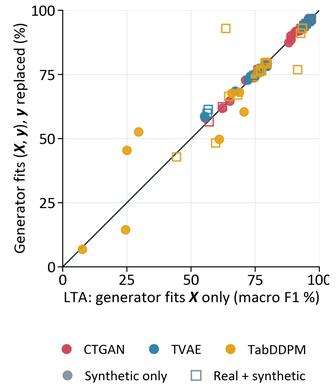


Regression (same layout; dashed line = real data only)



Open circle: the generator's own targets. Filled circle: LTA. Averaged over 2 seeds (LTA also over 3 labelers). Generated targets fall short of real data on 5 of 5 classification tasks for every generator; News has no predictable signal for any method, including real data ($|R^2| \le 0.03$).

Does the generator need *y*?



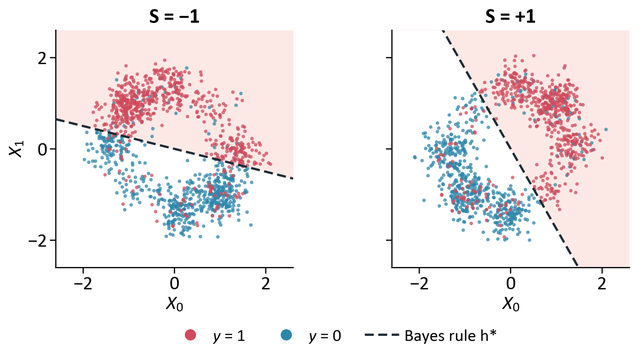
Each point is one task, generator and labeler. On the diagonal, fitting the generator to (***X***, ***y***) instead of ***X*** alone makes no difference.

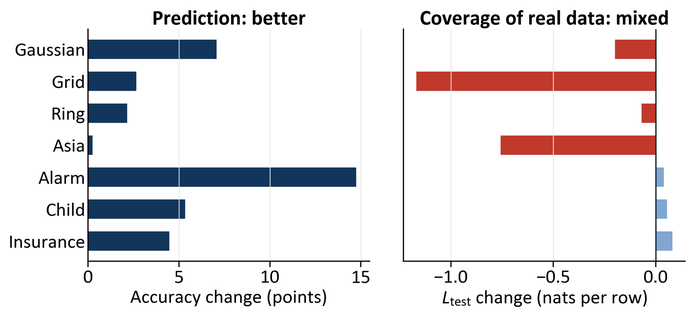
Joint minus LTA: **+0.5 points** (95% CI −3.0 to +3.9; Holm p = 1.00).

Fitting the generator to ***y*** as well adds no detectable benefit.

Simulated data with known truth

Ring: 8 Gaussian clusters in (X_0, X_1). A binary feature S sets the slope of the Bayes boundary h^*; 10% of labels are flipped. Six more datasets: Gaussian, Grid and 4 Bayesian networks.





L_{test}: fit a simple density model to the synthetic table, then average its log-density over real test rows. Higher means the synthetic table covers the real data better.

LTA raises accuracy on all 7 datasets (+0.3 to +14.7 points, within 1.2 of real data) but lowers L_{test} on 4: hard labels erase the target's natural randomness.